

Fundamentals of Digital Signal Processing: From Sampling to Phase-Vocoder Pitch Shifting

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Abstract

This report gives a self-contained mathematical overview of the core digital signal processing (DSP) concepts underlying modern audio analysis and manipulation tools: sampling and quantization, the Discrete and Fast Fourier Transforms, window functions, the Short-Time Fourier Transform (STFT), and four representative algorithms built on top of it — frequency-domain equalization, phase-vocoder pitch detection, and phase-vocoder pitch shifting. Each section develops the relevant mathematics from first principles and ends with an algorithmic summary suitable as a reference for implementation.

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1 Introduction

Digital signal processing is the of representing, analyzing, and manipulating signals — most commonly audio, but the same mathematics applies to images, radar, biomedical signals, and more — using discrete, finite-precision numbers rather than continuous physical quantities. Nearly every technique in modern audio software, from equalizers to pitch correction, rests on a small set of mathematical ideas: sampling theory, the Fourier transform, and the practical adaptations needed to apply Fourier analysis to signals that change over time. This report develops those ideas in order, building toward four concrete algorithms.

2 Fundamental Concepts

2.1 Continuous and Discrete Signals

A continuous-time signal $x_c(t)$ is defined over the observation interval $t \in [0;_d - 1]$. A discrete-time signal $x[n]$ where $n \in \{0, 1, 2, \dots, N - 1\}$, is obtained by sampling $x_c(t)$ at a uniform sampling interval Δt :

$$x[n] = x_c(n\Delta t), \quad f_s = \frac{1}{\Delta t} = \frac{N}{d}, \quad (1)$$

where f_s represents the sampling frequency expressed in hertz (Hz), T_d denotes the total duration of the signal in seconds, and N is the total number of discrete samples acquired.

2.2 The Sampling Theorem and Aliasing

The Nyquist–Shannon sampling theorem states that a continuous signal containing no frequency content above $f_{\max}$ can be perfectly reconstructed from its samples if and only if

$$f_s > 2f_{\max}. \quad (2)$$

The quantity $f_s/2$ is the **Nyquist frequency**. Sampling below this rate causes **aliasing**: frequency components above $f_s/2$ are indistinguishable from — and corrupt — components below it, folding back according to

$$f_{\text{alias}} = |f - kf_s|, \quad k \in \mathbb{Z} \text{ chosen so } f_{\text{alias}} \in [0, f_s/2]. \quad (3)$$

This is why real digitization systems place an analog anti-aliasing low-pass filter (AAF) before the sampler to block high-frequency components above the Nyquist limit before they can corrupt the digital data.

The standard sampling frequency $f_s = 44.1$ kHz was selected to align with the upper limit of human auditory perception (typically 20 Hz to 20 kHz). Under the Nyquist–Shannon sampling theorem, any sampling rate satisfying $f_s > 40$ kHz is mathematically sufficient to reconstruct human-audible signals without aliasing.

2.3 Quantization

Each sample is stored with finite precision — commonly b bits, representing 2^b discrete levels. This introduces quantization error, modeled as additive noise whose power determines the achievable **signal-to-quantization-noise ratio**:

$$\text{SQNR} \approx 6.02b + 1.76 \text{ dB}. \quad (4)$$

Heuristic. Suppose an Analog-to-Digital Converter (ADC) representing a signal using b bits and ADC's input range is $[-A, A]$. The distance between adjacent levels — the **quantization step** — is approximately

$$\Delta = \frac{2A}{2^b} = \frac{A}{2^{b-1}}$$

The quantization error is the difference between the original and quantized values. Assuming rounding to the nearest level, it lies approximately within

$$-\frac{\Delta}{2} \leq e \leq \frac{\Delta}{2}.$$

Modeling this error as uniformly distributed gives its noise power:

$$\text{Var}(e) = P_q = \frac{\Delta^2}{12}.$$

For a full-scale sinusoidal input $x(t) = A \sin(\omega t)$, the signal power is

$$P_s = \frac{A^2}{2}.$$

Therefore, the signal-to-quantization-noise ratio (SQNR) is

$$\text{SQNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_s}{P_q} \right) = 10 \log_{10}(3) + 10(2b - 1) \log_{10}(2) \approx 6.02b + 1.76 \text{ dB}.$$

□

Each additional bit of resolution buys roughly 6 dB of dynamic range — the origin of the common shorthand that 16-bit audio has about 96 dB of dynamic range.

2.4 The Decibel Scale

The decibel (dB) expresses a ratio on a logarithmic scale, reflecting that human perception of loudness (and many physical quantities) is approximately logarithmic. For power quantities P_1, P_2 :

$$L_{\text{dB}} = 10 \log_{10} \left(\frac{P_1}{P_2} \right). \quad (5)$$

Since power is proportional to amplitude squared, the equivalent amplitude-ratio form carries a factor of 20 rather than 10:

$$L_{\text{dB}} = 20 \log_{10} \left(\frac{A_1}{A_2} \right). \quad (6)$$

In audio, magnitudes are frequently expressed relative to a reference such as full-scale amplitude (**dBFS**) or a chosen threshold.

2.5 Complex Exponentials and Euler's Formula

Fourier analysis is dramatically simplified by representing sinusoids as complex exponentials via Euler's formulas:

$$e^{j\theta} = \cos \theta + j \sin \theta, \quad (7)$$

$$e^{-j\theta} = \cos \theta - j \sin \theta, \quad (8)$$

A real-valued sinusoid $A \cos(\omega t + \phi)$ can be written as the real part of a rotating **phasor** $A e^{j(\omega t + \phi)}$. Representing sinusoids in this complex form enables linear combinations—such as addition and scaling—of signals sharing the same frequency to be evaluated using standard complex arithmetic. Consequently, transformations like the Fourier transform are naturally complex-valued, even when operating on real-valued inputs. signals.

2.6 Energy, Power, and RMS

The energy of a discrete signal is $E = \sum_n |x[n]|^2$, and its average power over N samples is $P = \frac{1}{N} \sum_n |x[n]|^2$. The root-mean-square (RMS) value, $x_{\text{rms}} = \sqrt{P}$, is the standard measure of a signal's effective amplitude and the usual basis for loudness-related dB calculations.

3 The Discrete and Fast Fourier Transforms

3.1 From Continuous to Discrete Fourier Analysis

The continuous Fourier transform decomposes a signal into a continuum of complex exponentials. For a finite-length discrete signal, the analogous tool is the **Discrete Fourier Transform (DFT)**, which decomposes N samples into N discrete frequency components.

3.2 Definition

For a sequence $x[n]$, $n = 0, \dots, N-1$, the DFT is

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, \quad k = 0, \dots, N-1, \quad (9)$$

with inverse

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}. \quad (10)$$

Each output $X[k]$ is a complex number whose magnitude $|X[k]|$ and phase $\angle X[k]$ describe the amplitude and phase of the frequency component at

$$f_k = \frac{k f_s}{N}, \quad (11)$$

Heuristic. From Equation (9): We have Angular speed

$$w_k = \frac{2\pi k}{N}$$

$$f_k = w_k \frac{f_s}{2\pi} = \frac{k f_s}{N}$$

$$\Delta f = f_{k+1} - f_k = \frac{f_s}{N}$$

□

so the DFT gives a **frequency resolution** of $\Delta f = f_s/N$ — coarser for shorter transforms, a trade-off revisited in Section 5.

3.3 Key Properties

- **Linearity:** the DFT of a sum is the sum of the DFTs.
- **Periodicity:** the DFT implicitly treats $x[n]$ as one period of a periodic signal, which underlies both spectral leakage (Section 4) and the distinction between circular and linear convolution (Section 9.1).
- **Conjugate symmetry for real input:** if $x[n]$ is real, then $X[N - k] = X[k]^*$. This means half the spectrum ($k = 0, \dots, N/2$) contains all the information; any frequency-domain manipulation of a real signal must preserve this symmetry or the inverse transform will produce a signal with a spurious imaginary component.

Heuristic.

$$e^{-j2\pi \frac{N-k}{N}n} = e^{-j2\pi n} e^{j2\pi \frac{k}{N}n} = e^{j2\pi \frac{k}{N}n}$$

since $e^{-j2\pi n} = 1$ for all integer n . Evaluating $X[N - k]$ then gives:

$$X[N - k] = \sum_{n=0}^{N-1} x[n] e^{j2\pi \frac{k}{N}n} = X^*[k], \quad k = 0, \dots, N - 1.$$

Applying (7) and (8) confirms that $X[k]$ and $X[N - k]$ share identical real parts but possess imaginary components of opposite sign. Consequently, the magnitude spectrum is even-symmetric about $N/2$:

$$|X[k]| = |X[N - k]|.$$

□

3.4 The Fast Fourier Transform

Computing Equation (9) directly costs $O(N^2)$ multiply-adds. The **Fast Fourier Transform (FFT)**, most famously the Cooley–Tukey algorithm [1], reduces this to $O(N \log_2 N)$ by recursively splitting the DFT of length N (for N a power of two) into two DFTs of length $N/2$ over the even- and odd-indexed samples:

$$X[k] = \underbrace{\sum_{m=0}^{N/2-1} x[2m] e^{-j2\pi km/(N/2)}}_{E[k]} + e^{-j2\pi k/N} \underbrace{\sum_{m=0}^{N/2-1} x[2m+1] e^{-j2\pi km/(N/2)}}_{O[k]}. \quad (12)$$

The terms $e^{-j2\pi k/N}$ are called **twiddle factors**. Applying this split recursively down to *length* – 1 transforms, then combining results pairwise via “butterfly” operations $X[k] = E[k] + W_N^k O[k]$ and $X[k + N/2] = E[k] - W_N^k O[k]$, gives the full transform in $\log_2 N$ stages of $O(N)$ work each. The practical, iterative form of this algorithm reorders the input via a **bit-reversal permutation** so that each stage operates on contiguous, ever-larger blocks without recursion.

3.5 Inverse Discrete Fourier Transform and Inverse Fast Fourier Transform

To recover the discrete-time sequence $x[n]$ from its frequency-domain representation $X[k]$, we apply the **Inverse Discrete Fourier Transform (IDFT)**:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad n = 0, \dots, N-1. \quad (13)$$

Structurally, Equation (13) is nearly identical to the forward DFT, differing only by a positive sign in the complex exponential argument and a global normalization factor of $1/N$. Because of this profound symmetry, the identical divide-and-conquer strategy that yields the $O(N \log_2 N)$ FFT is directly applicable to the **Inverse Fast Fourier Transform (IFFT)**.

The IFFT simply replaces the forward twiddle factors $W_N^k = e^{-j2\pi k/N}$ with their complex conjugates $W_N^{-k} = e^{j2\pi k/N}$. In modern digital signal processing libraries, the IFFT is routinely computed by reusing the exact forward FFT subroutine. By exploiting conjugate symmetry, the inverse transform can be evaluated as:

$$x[n] = \frac{1}{N} (\text{FFT}(X[k]))^*, \quad (14)$$

where $(\cdot)^*$ denotes complex conjugation. This duality ensures that both forward and inverse operations share a unified hardware or software architecture, executing in identical $O(N \log_2 N)$ time.

4 Window Functions

4.1 Spectral Leakage

The DFT assumes the analyzed N -sample frame is one period of a periodic signal. If the true signal is not exactly periodic within that frame — true of essentially all real audio — the implicit discontinuity at the frame boundary introduces energy across many frequency bins that shouldn't be there. This is **spectral leakage**. Mathematically, truncating a signal to N samples is equivalent to multiplying by a rectangular window, and multiplication in time corresponds to convolution in frequency (Section 9.1) with the rectangular window's own spectrum — a sinc-like function with substantial side lobes.

4.2 Common Window Functions

Tapering the frame's edges to zero before transforming reduces the discontinuity and hence leakage, at the cost of widening the main lobe (reducing frequency resolution). Commonly used windows ($n = 0, \dots, N-1$):

$$\text{Rectangular: } w[n] = 1 \quad (15)$$

$$\text{Hann: } w[n] = 0.5 \left(1 - \cos \frac{2\pi n}{N} \right) \quad (16)$$

$$\text{Hamming: } w[n] = 0.54 - 0.46 \cos \frac{2\pi n}{N} \quad (17)$$

$$\text{Blackman: } w[n] = 0.42 - 0.5 \cos \frac{2\pi n}{N} + 0.08 \cos \frac{4\pi n}{N} \quad (18)$$

Table 1: Approximate trade-offs of common windows [2].

Window	Main-lobe width	Peak side-lobe level
Rectangular	2 bins	−13 dB
Hann	4 bins	−31 dB
Hamming	4 bins	−43 dB
Blackman	6 bins	−58 dB

4.3 Periodic vs. Symmetric Windows

The formulas above use denominator N (the *periodic* form). An alternative, more common in filter-design contexts, uses denominator $N - 1$ (the *symmetric* form). The periodic form is specifically required for overlap-add reconstruction to work exactly (Section 5); using the symmetric form there introduces a small periodic amplitude ripple.

5 The Short-Time Fourier Transform (STFT)

5.1 Motivation

The DFT assumes stationarity: a single frequency content for the entire analyzed frame. Real audio signals are highly non-stationary — pitch, timbre, and loudness all change over time. The **Short-Time Fourier Transform** addresses this by applying the DFT to a sliding, windowed segment of the signal, producing a time-varying spectrum.

5.2 Definition

Given a window $w[n]$ of length N and a **hop size** H (the number of samples advanced between successive frames), the STFT is

$$X[m, k] = \sum_{n=0}^{N-1} x[n + mH] w[n] e^{-j2\pi kn/N}, \quad (19)$$

where m indexes the frame (time) and k indexes the frequency bin. The ratio N/H is the **overlap factor** ($H = N/2$ is “50% overlap”).

5.3 Time-Frequency Resolution Trade-off

A shorter window gives better time resolution but worse frequency resolution ($\Delta f = f_s/N$), and vice versa — a manifestation of the time-frequency uncertainty principle, $\Delta t \cdot \Delta f \gtrsim \text{constant}$. There is no window length that is simultaneously optimal for both; the choice depends on the application (e.g., transient detection favors short windows, precise pitch estimation favors long ones).

5.4 Inversion: the COLA Constraint and the Weighted Overlap-Add

Reconstructing a time-domain signal from a Short-Time Fourier Transform (STFT) involves applying the Inverse Discrete Fourier Transform (IDFT) to each frame, followed by an **Overlap-Add (OLA)** procedure. Each frame’s inverse transform is shifted by the hop size H and summed into an output buffer. In the identity case (no spectral modification), the reconstructed signal $y[n]$ is given by:

$$y[n] = \frac{\sum_m y_m[n - mH]}{\sum_m w[n - mH]}. \quad (20)$$

For exact reconstruction without local normalization, the analysis window $w[n]$ must satisfy the **Constant Overlap-Add (COLA) constraint**:

$$\sum_{m=-\infty}^{\infty} w[n - mH] = c, \quad \text{for all } n \in [1, N]. \quad (21)$$

A periodic Hann window at $H = N/2$ satisfies this with $c = 1$, meaning overlapping windowed frames sum to exactly the original signal with no further normalization needed.

Heuristic. For $H = N/2$, $\sum_{m=-\infty}^{\infty} w[n - mH]$ becomes:

$$\sum_{m=0}^1 w[n - m\frac{N}{2}] = w[n] + w[n - \frac{N}{2}] = 1 - 0.5(\cos(2\pi\frac{n}{N}) + \cos(2\pi\frac{n - \frac{N}{2}}{N})) = 1$$

□

Alternatively, the Weighted Overlap-Add (WOLA) method applies windowing at both the analysis and synthesis stages. Modern DSP applications often prefer WOLA because applying a synthesis window $w_s[n] = w[n]$ smooths frame boundaries and suppresses spectral artifacts introduced during non-linear STFT modifications. The output signal is normalized by the local sum of squared window values:

$$y[n] = \frac{\sum_m w[n - mH] y_m[n - mH]}{\sum_m w^2[n - mH]}. \quad (22)$$

For exact reconstruction under WOLA without division by a time-varying denominator, the window must satisfy the squared-COLA constraint:

$$\sum_{m=-\infty}^{\infty} w^2[n - mH] = c, \quad \text{for all } n \in [1, N]. \quad (23)$$

Using the power-reduction identity, the squared Hann window expands to:

$$w^2[n] = \left[0.5 - 0.5 \cos\left(\frac{2\pi n}{N}\right)\right]^2 = \frac{3}{8} - \frac{1}{2} \cos\left(\frac{2\pi n}{N}\right) + \frac{1}{8} \cos\left(\frac{4\pi n}{N}\right). \quad (24)$$

Thus, a periodic Hann window at $H = N/4$ satisfies the squared-COLA condition with $c = 3/2$, guaranteeing perfect reconstruction under WOLA up to a constant scaling factor.

Heuristic.

$$\sum_{m=-\infty}^{\infty} (w^2[n - mH]) = \sum_{m=-\infty}^{\infty} \left(\frac{3}{8} - \frac{1}{2} \cos\left(\frac{2\pi(n - mH)}{N}\right) + \frac{1}{8} \cos\left(\frac{4\pi(n - mH)}{N}\right)\right)$$

For a 75% overlap ($H = N/4$), evaluating the sum over $m \in \{0, 1, 2, 3\}$ yields:

$$\sum_{m=0}^3 w^2[n - mH] = \frac{3}{2} - \frac{1}{2} \sum_{m=0}^3 \cos\left(\frac{2\pi n}{N} - \frac{\pi m}{2}\right) + \frac{1}{8} \sum_{m=0}^3 \cos\left(\frac{\pi n}{N} - \pi m\right) = \frac{3}{2}, \quad \text{for all } n \in [1; N]$$

□

5.5 Weighted Overlap-Add for Arbitrary Hops

When the synthesis hop size is arbitrary (notably pitch shifting algorithm, Section 8), the overlap between adjacent frames is generally not constant. Therefore, a fixed normalization factor may not be sufficient. A robust approach is to use **Weighted Overlap-Add (WOLA)** with a sample-dependent normalization factor.

Let $w_a[n]$ and $w_s[n]$ denote the analysis and synthesis windows, respectively. After processing frame m in the frequency domain and applying the synthesis window, the reconstructed signal is $\sum_m w_s[n - mH_s] y_m[n - mH_s]$, where H_s is the synthesis hop size.

The effective window contribution at each output sample is $\sum_m w_a[n - mH_s] w_s[n - mH_s]$.

Therefore, the reconstructed signal is normalized as $\frac{\sum_m w_s[n - mH_s] y_m[n - mH_s]}{\sum_m w_a[n - mH_s] w_s[n - mH_s]}$

For the common case where the same window is used for analysis and synthesis,

$$w_a[n] = w_s[n] = w[n],$$

$$\text{the normalization becomes } \frac{\sum_m w[n - mH_s] y_m[n - mH_s]}{\sum_m w^2[n - mH_s]}$$

Unlike a fixed scale factor, this normalization adapts to the actual overlap at every sample. It is therefore suitable for arbitrary synthesis hop sizes, provided that $\sum_m w^2[n - mH_s] > 0$ for every reconstructed sample.

For special window/hop combinations satisfying a constant-overlap-add condition, the denominator is constant and can be replaced by a single scale factor. However, for arbitrary hops, maintaining a per-sample normalization buffer is the more robust approach.

6 Algorithm: Frequency-Domain Equalization

6.1 Principle

Filtering — boosting or attenuating specific frequency ranges — can be performed either as time-domain convolution with a filter's impulse response, or, by the convolution theorem (Section 9.1), as simple multiplication in the frequency domain. An STFT-based equalizer exploits this directly.

6.2 Algorithm

Algorithm 1 STFT-based multi-band equalizer

```

1: for each frame  $m$  do
2:   Compute  $X[m, k] \leftarrow$  STFT frame (window + FFT)
3:   for  $k = 0$  to  $N/2$  do
4:      $f \leftarrow k \cdot f_s/N$ 
5:      $g \leftarrow$  gain for the band containing  $f$  (e.g., low/mid/high)
6:      $X[m, k] \leftarrow g \cdot X[m, k]$ 
7:     if  $k \neq 0$  and  $k \neq N/2$  then
8:        $X[m, N - k] \leftarrow g \cdot X[m, N - k]$  ▷ mirror: preserve conjugate symmetry
9:     end if
10:  end for
11:   $y_m \leftarrow$  IFFT( $X[m, \cdot]$ )
12:  Overlap-add  $y_m$  into the output at hop offset  $mH$ 
13: end for

```

6.3 The Conjugate Symmetry Constraint

Because the input is real, its spectrum is conjugate-symmetric (Section 3.3). Any per-bin gain g_k applied to a real signal’s spectrum *must* satisfy $g_{N-k} = g_k$ (a real gain, mirrored identically), or the inverse transform will produce a signal with a non-negligible imaginary part — i.e., it silently stops being valid real-valued audio. This is a common and easy-to-miss implementation bug.

6.4 Circular Convolution Artifacts

Multiplying an N -point spectrum is equivalent, in the time domain, to *circular* convolution of the windowed frame with the gain curve’s implied impulse response — not linear convolution. When the effective impulse response is short relative to the frame length N (true for smooth equalization curves), the wraparound error is negligible. Sharp, narrow frequency-domain features (a single bin zeroed abruptly, for instance) imply a long impulse response and can introduce audible ringing — addressed properly by the overlap-add/overlap-save *filtering* methods (Section 9.1).

7 Algorithm: Phase Vocoder Pitch Detection

7.1 The Resolution Problem

The DFT’s frequency resolution is $\Delta f = f_s/N$: at $f_s = 44100$ Hz and $N = 2048$, roughly 21.5 Hz per bin — far too coarse to resolve pitch to musically meaningful precision. Simply picking the bin with maximum magnitude cannot achieve the sub-bin accuracy needed for tuning applications.

7.2 Instantaneous Frequency from Phase

The key insight, central to the **phase vocoder** [3, 4], is that *phase* carries additional frequency information beyond what the bin index alone provides. A sinusoid sitting exactly at a bin’s center frequency $f_k = kf_s/N$ has a perfectly predictable phase advance over one hop of H samples:

$$\Delta\phi_{\text{expected}}[k] = \frac{2\pi kH}{N}. \quad (25)$$

Comparing the *actual* measured phase advance between consecutive STFT frames to this expectation reveals how far the true frequency deviates from the bin center. Let $\phi_m[k]$ be the phase of bin k in frame m . Define the phase deviation, wrapped to the principal value $(-\pi, \pi]$:

$$\Delta\phi[k] = \text{wrap}\left(\phi_m[k] - \phi_{m-1}[k] - \Delta\phi_{\text{expected}}[k]\right). \quad (26)$$

The refined, **instantaneous frequency** estimate for bin k is then

$$\hat{f}[k] = f_k + \frac{\Delta\phi[k] \cdot f_s}{2\pi H}. \quad (27)$$

This estimator gives accuracy far finer than the bin spacing alone would suggest, limited mainly by numerical precision rather than Δf .

7.3 Algorithm

Algorithm 2 Phase-vocoder pitch detection

```

1: Maintain  $\phi_{\text{prev}}[k]$  from the previous frame (init. to 0)
2: for each new hop of  $H$  samples do
3:   Compute windowed FFT  $\rightarrow X[k]$ ,  $k = 0, \dots, N/2$ 
4:   for each bin  $k$  do
5:      $\text{mag}[k] \leftarrow |X[k]|$ ,  $\phi[k] \leftarrow \angle X[k]$ 
6:     Compute  $\hat{f}[k]$  via Equation (27)
7:      $\phi_{\text{prev}}[k] \leftarrow \phi[k]$ 
8:   end for
9:    $k^* \leftarrow \arg \max_k \text{mag}[k]$  subject to  $f_{\min} \leq \hat{f}[k] \leq f_{\max}$ 
10:  if  $\text{mag}[k^*]$  exceeds a confidence threshold then report  $\hat{f}[k^*]$ 
11:  else report “no confident pitch” (silence/noise)
12: end for
```

7.4 Alternative Pitch Detection Methods

The phase vocoder approach is one of several standard techniques:

- **Autocorrelation:** finds the lag τ maximizing $R[\tau] = \sum_n x[n]x[n + \tau]$; the fundamental period is the first strong peak. Simple and effective, but sensitive to formants/harmonics that can cause octave errors.
- **Cepstral method:** computes $c[n] = \text{IFFT}(\log |\text{FFT}(x)[k]|)$; periodicity in the log-spectrum (harmonics) appears as a peak in the *quefrency* domain at the fundamental period.
- **YIN** [6]: an improved difference-function method addressing common autocorrelation failure modes, widely used in music information retrieval.

Each has different robustness/complexity trade-offs; the phase vocoder method is attractive when an STFT pipeline already exists for other purposes (e.g., alongside equalization or pitch shifting).

8 Algorithm: Phase Vocoder Pitch Shifting

8.1 Decoupling Pitch and Duration

Naively resampling a signal changes both its pitch and its duration together (playing a recording faster raises its pitch and shortens it). Musically useful pitch shifting requires changing pitch *without* changing duration. The classic two-stage solution:

1. **Time-stretch** the signal by ratio α using a phase vocoder — this changes duration but, crucially, preserves pitch, because the phase vocoder explicitly reconstructs the correct instantaneous frequencies regardless of hop size.
2. **Resample** the stretched result by $1/\alpha$. Resampling *does* link pitch and speed, so this restores the original duration while scaling the pitch by α .

8.2 Time-Stretching via Phase Accumulation

Let R_a be the analysis hop and $R_s = \text{round}(\alpha R_a)$ the synthesis hop. For each analysis frame, compute magnitude and instantaneous frequency $\hat{f}[k]$ exactly as in Section 7. Rather than reuse the measured phase directly, *synthesize* a new phase track that advances at the correct instantaneous frequency but over the (different) synthesis hop duration:

$$\psi_m[k] = \psi_{m-1}[k] + \hat{f}[k] \cdot \frac{2\pi R_s}{f_s}. \quad (28)$$

The synthesis spectrum for the frame is then reassembled from the *original* magnitude with the *new* phase:

$$X'_m[k] = |X_m[k]| e^{j\psi_m[k]}. \quad (29)$$

Each frame is inverse-transformed and overlap-added at hop R_s using WOLA (Section 5), since R_s is generally not a COLA-friendly hop size.

8.3 Resampling

The time-stretched signal (length $\approx \alpha$ times the original) is resampled back to the original length by evaluating it at positions scaled by α :

$$y[n] = \text{interp}(y_{\text{stretched}}, n \cdot \alpha), \quad (30)$$

using interpolation (linear, or higher-order/sinc-based for better fidelity) since $n\alpha$ generally falls between integer sample indices.

8.4 Algorithm

Algorithm 3 Phase-vocoder pitch shifting by ratio α

- 1: $R_s \leftarrow \text{round}(\alpha R_a)$
 - 2: **for** each analysis frame at hop R_a **do**
 - 3: Compute windowed FFT, magnitude $|X[k]|$, instantaneous frequency $\hat{f}[k]$ (Eq. 27)
 - 4: Accumulate synthesis phase $\psi[k]$ via Equation (28)
 - 5: $X'[k] \leftarrow |X[k]| e^{j\psi[k]}$, mirrored for conjugate symmetry
 - 6: IFFT X' , apply synthesis window, overlap-add at hop R_s with WOLA normalization
 - 7: **end for**
 - 8: Resample the resulting (time-stretched) signal by $1/\alpha$ to restore original duration
-

8.5 Known Limitations

Reusing magnitude while synthesizing a new phase track can smear transients (attacks lose sharpness — sometimes called “phasiness”) and can desynchronize the phase relationships between bins that were originally locked together (e.g., harmonics of the same note), producing a reverberant artifact. **Phase locking** techniques [5] address this by constraining phase relationships between neighboring bins during resynthesis, at the cost of additional complexity.

8.6 Improved Approach: Direct Spectral Pitch Shifting

The previous approach performs pitch shifting indirectly through two separate operations: time-stretching followed by resampling. Although this method is conceptually simple, the resampling stage is unnecessary when the desired pitch transformation can be performed directly in the frequency domain.

The improved implementation therefore performs **direct spectral pitch shifting** within the phase-vocoder framework. For a pitch-shift ratio α , an output frequency bin k corresponds approximately to the input bin

$$k_{\text{src}} = \frac{k}{\alpha}. \quad (31)$$

Thus, instead of changing the synthesis hop size, the spectral components are remapped to their new frequency locations. The magnitude at the source position is interpolated to avoid abrupt changes caused by integer bin mapping:

$$|Y[k]| = \text{interp}(|X[k_{\text{src}}]|). \quad (32)$$

The instantaneous frequency estimated from the source bin is also scaled by α . In bin units,

$$\hat{f}_{\text{out}}[k] = \alpha \hat{f}_{\text{in}}[k_{\text{src}}] \quad (33)$$

The synthesis phase is then accumulated across frames using the shifted frequency:

$$\psi_m[k] = \psi_{m-1}[k] + 2\pi \hat{f}_{\text{out}}[k] \frac{H}{f_s}, \quad (34)$$

where H is the fixed analysis/synthesis hop size and N is the FFT size. The resulting spectrum is reconstructed as

$$Y_m[k] = |Y_m[k]| e^{j\psi_m[k]}. \quad (35)$$

After the inverse FFT, the frames are windowed and reconstructed using WOLA. Unlike the previous implementation, the hop size remains fixed, so the output has the same duration as the input without requiring a separate resampling stage.

8.6.1 Time-Varying Pitch Correction

A further advantage of the direct approach is that the pitch-shift ratio does not need to be constant throughout the signal. For frame m , the desired correction can be specified by a semitone value s_m , giving

$$\alpha_m = 2^{s_m/12}. \quad (36)$$

The pitch shifter can therefore process an entire audio signal using a frame-by-frame correction track. This is useful for autotune, where the required correction changes over time according to the detected pitch.

To avoid abrupt changes between adjacent frames, the correction track is interpolated onto the pitch-shifter frame grid and smoothed before resynthesis. The phase state is maintained continuously across all frames, allowing the pitch correction to change gradually without introducing independent phase discontinuities between blocks.

8.6.2 Comparison with the Original Approach

The two approaches differ mainly in how the pitch transformation is performed:

- **Original:** time-stretch using a phase vocoder, then resample to restore the original duration.
- **Improved:** directly remap spectral frequencies and synthesize the shifted phase, keeping the duration unchanged.
- **Original:** primarily designed around a single, constant pitch-shift ratio.
- **Improved:** naturally supports a time-varying pitch-shift track, making it more suitable for autotune.
- **Both:** rely on instantaneous-frequency estimation, phase accumulation, IFFT, and overlap-add reconstruction.

The improved method removes the additional resampling stage and provides more direct control over the pitch transformation. However, it does not eliminate the fundamental limitations of phase-vocoder processing. Transient smearing and phase incoherence between related spectral components can still occur. More advanced techniques such as phase locking, transient detection, and peak-based spectral processing can be introduced to address these artifacts.

9 Additional Topics

9.1 The Convolution Theorem

Convolution in one domain corresponds to multiplication in the other:

$$x[n] * h[n] \iff X[k] \cdot H[k]. \quad (37)$$

This is the theoretical foundation for all frequency-domain filtering (Section 6): an FIR filter can be applied either by direct time-domain convolution or by multiplying spectra. Because the DFT implicitly treats sequences as periodic, direct spectral multiplication computes *circular* convolution;

$$IFFT(X[k] H[k]) = x[n] \circledast h[n]. \quad (38)$$

when *linear* convolution is required (the usual case for filtering a finite signal), the frames must be zero-padded to at least $N_x + N_h - 1$ samples before transforming.

$$IFFT[FFT(x_{pad}) FFT(h_{pad})] = x * h. \quad (39)$$

The **overlap-add** and **overlap-save** methods apply this idea block-by-block to filter arbitrarily long signals efficiently.

9.2 The Uncertainty Principle in Time-Frequency Analysis

No time-frequency representation can achieve arbitrarily fine resolution in both domains simultaneously; for a Gaussian window this bound is exact and analogous to the Heisenberg uncertainty principle. This motivates alternative representations when a single STFT resolution is unsuitable across the full frequency range of interest.

9.3 Beyond the STFT: Constant-Q and Wavelet Transforms

The STFT uses a fixed window length, giving uniform frequency resolution ($\Delta f = f_s/N$ everywhere) but non-uniform resolution in *musical* terms, since perceived pitch is logarithmic. The **Constant-Q Transform** and **wavelet transforms** use frequency-dependent window lengths (longer windows at low frequencies, shorter at high frequencies) to give roughly constant relative resolution — better matched to music analysis, at the cost of no longer having a simple fixed-size FFT implementation.

10 Conclusion

A remarkably small set of ideas — sampling theory, the DFT/FFT, windowing, and the STFT’s analysis-synthesis framework — underlies a wide range of practical audio algorithms. Frequency-domain equalization follows directly from the convolution theorem; pitch detection and pitch shifting both hinge on extracting more precise frequency information from phase than magnitude alone can provide, via the phase vocoder’s instantaneous frequency estimator. Understanding these foundations makes it possible to reason about, debug, and extend a large fraction of real-world audio DSP software.

References

- [1] J. W. Cooley and J. W. Tukey, *An Algorithm for the Machine Calculation of Complex Fourier Series*, Mathematics of Computation, vol. 19, pp. 297–301, 1965.
- [2] F. J. Harris, *On the Use of Windows for Harmonic Analysis with the Discrete Fourier Transform*, Proceedings of the IEEE, vol. 66, no. 1, pp. 51–83, 1978.
- [3] J. L. Flanagan and R. M. Golden, *Phase Vocoder*, Bell System Technical Journal, vol. 45, pp. 1493–1509, 1966.
- [4] M. Dolson, *The Phase Vocoder: A Tutorial*, Computer Music Journal, vol. 10, no. 4, pp. 14–27, 1986.
- [5] J. Laroche and M. Dolson, *Improved Phase Vocoder Time-Scale Modification of Audio*, IEEE Transactions on Speech and Audio Processing, vol. 7, no. 3, pp. 323–332, 1999.
- [6] A. de Cheveigné and H. Kawahara, *YIN, a Fundamental Frequency Estimator for Speech and Music*, Journal of the Acoustical Society of America, vol. 111, no. 4, pp. 1917–1930, 2002.
- [7] A. V. Oppenheim and R. W. Schaffer, *Discrete-Time Signal Processing*, 3rd ed., Prentice Hall, 2009.
- [8] J. O. Smith III, *Spectral Audio Signal Processing*, online book, <https://ccrma.stanford.edu/~jos/sasp/>, accessed 2026.

- [9] C. E. Shannon, *Communication in the Presence of Noise*, Proceedings of the IRE, vol. 37, no. 1, pp. 10–21, 1949.